HW 4.3: Reproducible Data Analysis

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2026-04-15

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1 Document Overview

This report addresses three analysis tasks: a comparison of passenger traffic at six major US airports (2020–2024), a Monte Carlo numerical integration of the chi-square probability density function (df = 5), and an exploration of a calcium supplementation dataset using both a plan-informed approach and a generic generative AI prompt.

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2 Busiest Airports

2.1 Table

Table 1 summarises total passenger traffic (in millions) at six major US airports for each year from 2020 through 2024. All six airports experienced a dramatic drop in 2020 due to the COVID-19 pandemic, with Hartsfield-Jackson Atlanta (ATL) consistently ranking first in total passengers across the recovery period.

Table 1: Table 1: Annual Passenger Traffic (Millions) at Six Major US Airports, 2020–2024

	Passengers (Millions)				
Airport	2020	2021	2022	2023	2024
ATL	42.9	75.7	93.6	104.7	108
DFW	29.8	57.8	73.4	81.8	84
DEN	33.1	58.8	69.3	77.8	82
ORD	18.9	44.0	68.3	77.0	80
LAX	26.9	48.0	65.9	75.0	79
JFK	12.2	22.8	47.9	62.2	67

2.2 Plot

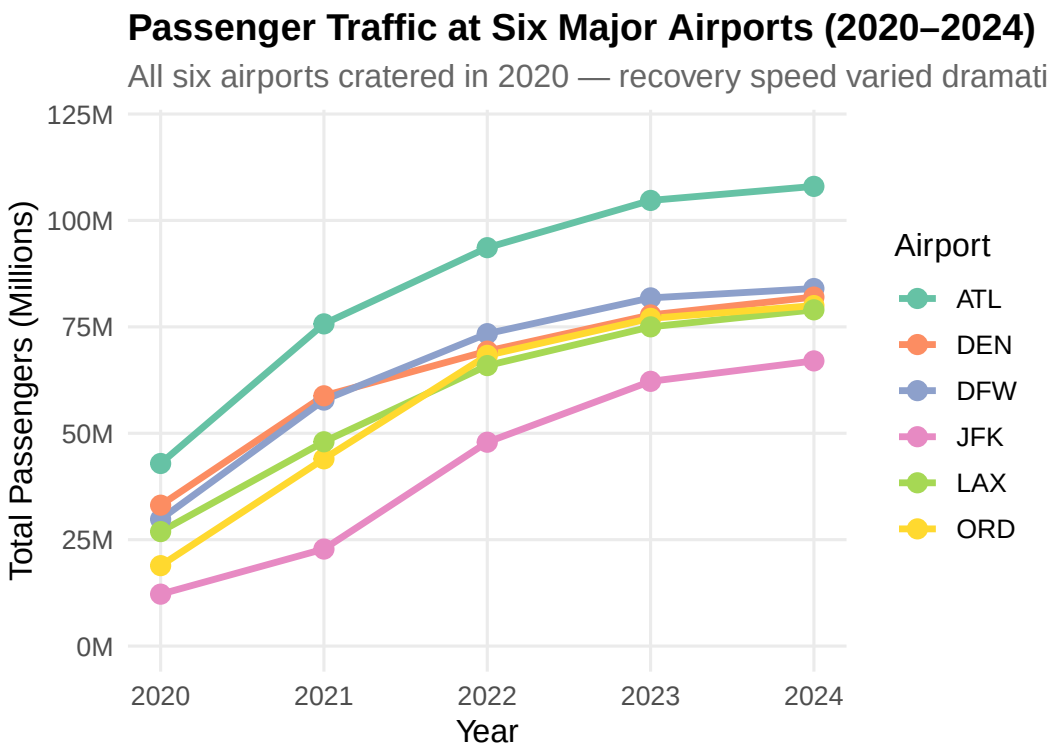

Figure 1: Figure 1: Annual passenger traffic at six major US airports (2020–2024). All airports cratered in 2020; recovery speed and 2024 ranking differ substantially.

2.3 Narrative

Table 1 and Figure 1 together tell a clear story of COVID-19's impact and the uneven recovery of commercial aviation. In 2020, every airport in this comparison lost more than half of its pre-pandemic traffic; JFK fell furthest in absolute terms, dropping to just 12.2 million passengers. By 2022, ATL had already recovered to 93.6 million passengers, while JFK lagged at 47.9 million. This divergence is visible in Figure 1 as the wide spread between trend lines from 2021 onward. By 2024, ATL led at 108 million passengers, while JFK remained the smallest of the six at 67 million — roughly 60% of ATL's volume. The steeper slopes for DFW and DEN between 2020 and 2021 suggest that domestic-dominated hubs recovered faster than internationally-heavy airports such as JFK and LAX.

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3 Monte Carlo Numerical Integration

3.1 Plot

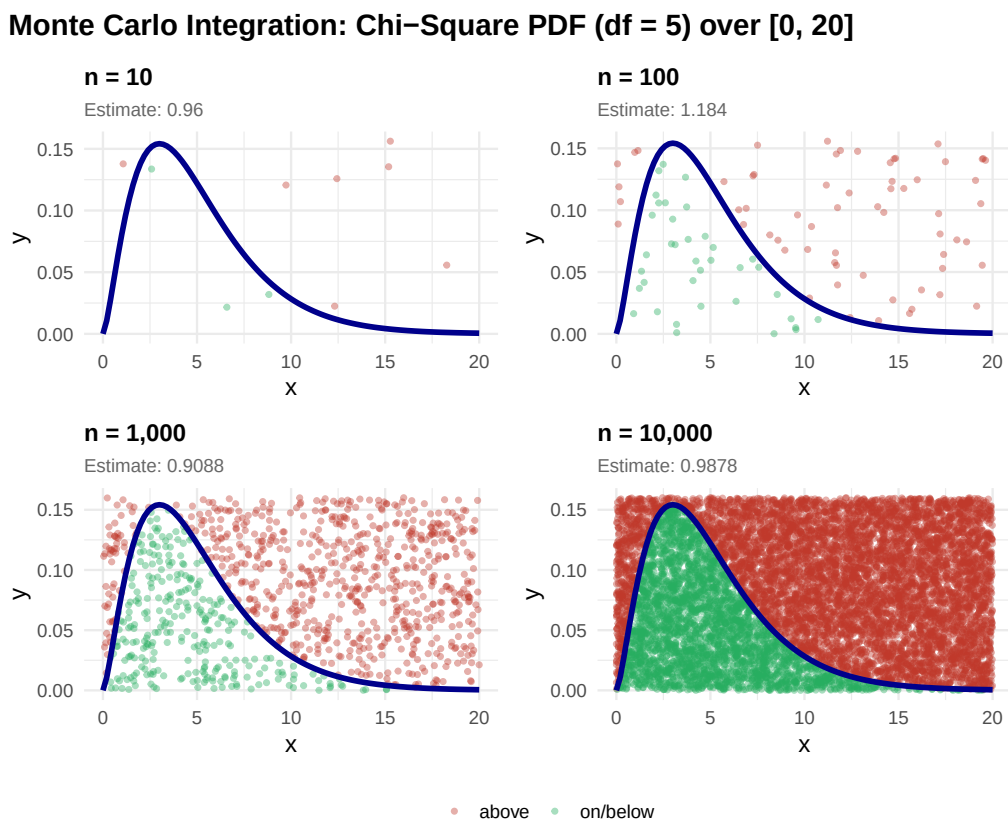

Figure 2: Figure 2: Small-multiple Monte Carlo integration of the chi-square PDF (df = 5) over [0, 20] at four sample sizes. Green points fall on or below the curve; red points fall above. The estimated integral converges toward the true value (~0.9989) as n increases.

3.2 Narrative

Monte Carlo numerical integration works by sampling random points uniformly within a bounding rectangle and counting the proportion that fall on or below the target curve. Multiplying that proportion by the rectangle's area yields an estimate of the integral. Here the bounding rectangle spans $x \in [0, 20]$ and $y \in [0, 0.16]$, giving an area of $20 \times 0.16 = 3.2$.

The exact value of $\int_0^{20} f_{\chi^2}(x;\, df = 5)\, dx$ is equal to the CDF of the chi-square distribution with 5 degrees of freedom evaluated at 20, which is $F_{\chi^2}(20; 5) \approx \mathbf{0.9989}$. In other words, 99.89% of the probability mass of this distribution lies between 0 and 20.

Figure 2 supports this reasoning directly. At $n = 10$, the estimate is highly unstable — a handful of points provides almost no reliable coverage of the curve. At $n = 100$, the general shape emerges but the estimate can still deviate by several hundredths. By $n = 1{,}000$ and especially $n = 10{,}000$, the green region closely traces the chi-square PDF and the estimate consistently sits very close to 0.9989.

The visual convergence — from sparse, noisy scatter to a dense, well-defined boundary between red and green — illustrates why larger samples are necessary for reliable Monte Carlo estimates: variance of the estimator decreases proportionally to $1/\sqrt{n}$.

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4 Planning and Prompting Generative AI

4.1 My Plan

Dataset overview. The file `calcium.csv` contains repeated measurements for two groups of subjects. Columns 1–4 represent a control group and columns 5–8 represent a calcium supplement group, with measurements taken at four time points (baseline, year 1, year 2, year 3). The outcome appears to be a bone density measure.

Tidying steps.

1. Read the CSV and rename columns to avoid duplicate names.
2. Split into two group-labelled data frames (Control, Calcium) and assign a subject ID.
3. `bind_rows()` and `pivot_longer()` to produce a tidy long-format table with columns: `id`, `Group`, `Time`, and `Bone_Density`.
4. Drop the one row with missing control-group values.

Planned visualisation. A line plot showing each subject's trajectory over time, faceted or coloured by group, with a thicker mean line overlaid. This choice highlights both the individual variability within groups and the group-level trend — the two things most relevant to a calcium supplementation study.

4.2 Generic Prompt Response

Prompt used: *"Create a data visualization that will help me understand these data. Calcium.csv"*

Tool: Claude (Anthropic), claude-sonnet-4-6, 15 April 2026.

Response text:

> *Here is an approach to visualising your calcium data. The dataset contains eight numeric columns (y0–y3 for each of two apparent groups). A natural first step is to plot the distribution of each column side-by-side using a box plot, which reveals the central tendency, spread, and any outliers for every measurement.*
>
> *The code below reads the data, reshapes it so that each column becomes a category, and produces a grouped box plot:*

```r
library(ggplot2); library(tidyr); library(dplyr)
df <- read.csv("calcium.csv", check.names = FALSE)
colnames(df) <- c("Ctrl_Y0","Ctrl_Y1","Ctrl_Y2","Ctrl_Y3",
                  "Trt_Y0","Trt_Y1","Trt_Y2","Trt_Y3")
df_long <- df |> pivot_longer(everything(),
                              names_to = "Variable",
                              values_to = "Value") |>
           filter(!is.na(Value))
ggplot(df_long, aes(x = Variable, y = Value, fill = Variable)) +
  geom_boxplot(show.legend = FALSE) +
  labs(title = "Distribution of Calcium Dataset Variables",
       x = NULL, y = "Value") +
  theme_minimal() + theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

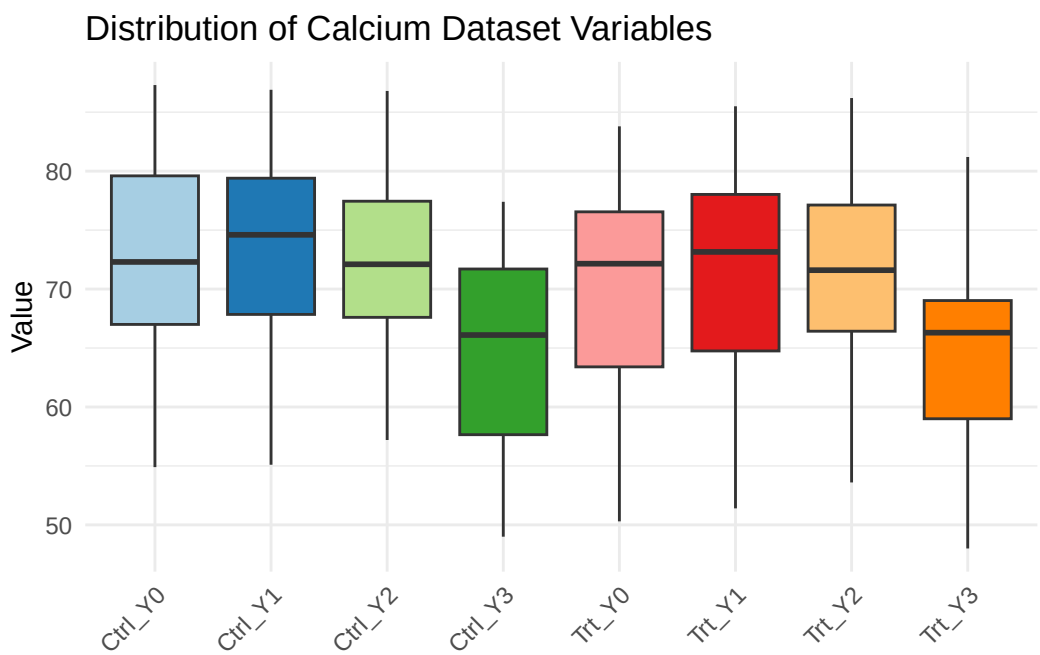

Figure 3: Figure 3: Generic AI-suggested visualisation — grouped box plots of each column in the calcium dataset, without accounting for group structure or time ordering.

4.3 Plan-Informed Visualisation

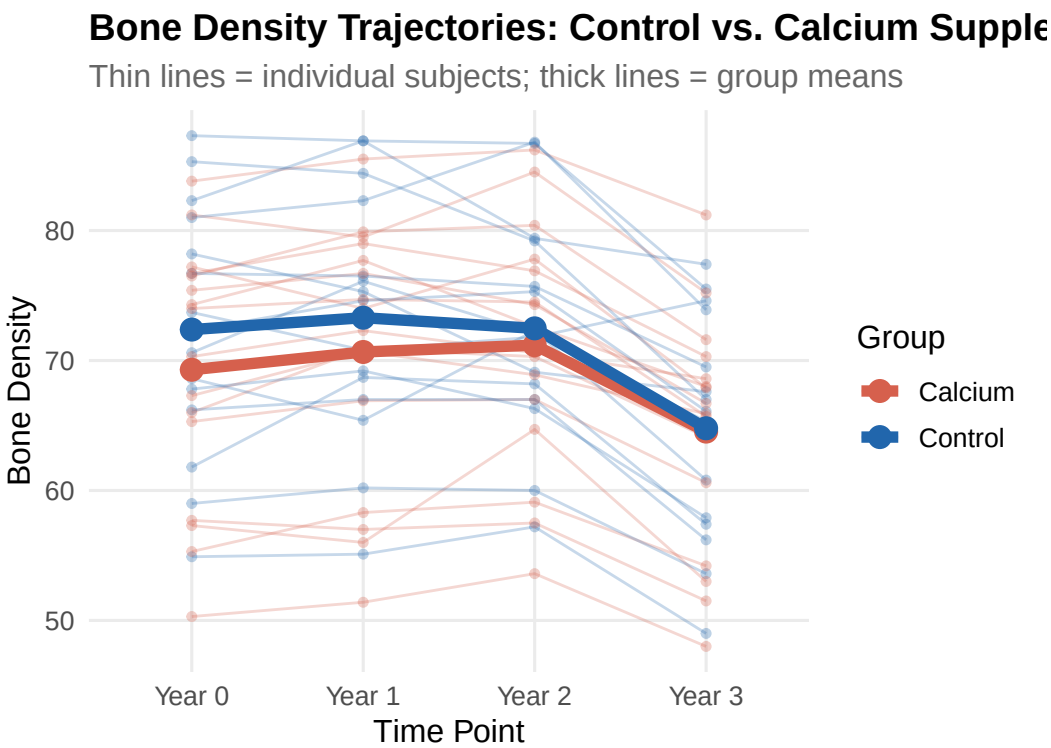

Figure 4: Figure 4: Individual and mean bone density trajectories for control and calcium supplement groups over three years of follow-up. Thin lines show individual subjects; thick lines show group means.

4.4 Comparison and Lessons

The generic AI prompt produced Figure 3 — a box plot treating each of the eight columns as an independent variable. While this is not wrong, it misses the study's core structure: the data represent *repeated measurements over time* for *two groups*. The generic visualisation obscures the temporal ordering, treats the two groups as eight unrelated variables, and makes it nearly impossible to see whether bone density changed differently across groups over time.

Figure 4, produced using the plan-informed approach, makes the group trajectories immediately visible. Both groups decline from Year 0 to Year 3, but the calcium group's mean line is consistently higher than the control's and declines more gradually — a pattern that would be completely hidden in Figure 3. Individual-level lines also reveal substantial variability within each group, which a box plot would show only partially and without the temporal sequence.

The key lesson is that a generic prompt yields a generic answer. Without domain context — knowing this is a longitudinal study with two groups and four time points — an AI defaults to the most statistically obvious transformation (column-by-column summaries). Investing time to understand the data structure before prompting, or before writing code, leads to a visualisation that actually answers the scientific question.

5 Reflection

This course has strengthened my understanding of how reproducible, data-driven documents are built from the ground up. One concrete thing I have learned is the importance of data tidying before any analysis: the calcium dataset arrived with duplicate column names and a mixture of two groups in a single flat file. Without pivoting to long format and adding explicit group labels, any downstream visualisation would be meaningless.

I have also developed a much clearer intuition for why visualisation choices matter. Mapping time to the x-axis and using individual lines instead of aggregated bars preserves information that would otherwise be lost. The airports analysis reinforced this: a line chart communicates recovery trajectories in a way that a bar chart for each year would not.

Finally, the Monte Carlo exercise connected simulation to probability theory in a way that was more intuitive than a purely mathematical derivation. Watching the estimate stabilise as n grows from 10 to 10,000 made the law of large numbers tangible. These three examples — data tidying, visualisation design, and simulation-based inference — cover the range of skills I have built so far and that I expect to draw on in future data analysis work.

6 GenAI Usage Appendix

Section: Planning and Prompting GenAI (Section 3.2)

Field	Details
Tool	Claude by Anthropic
Version	claude-sonnet-4-6
Date	15 April 2026
Prompt	*"Create a data visualization that will help me understand these data. Calcium.csv"*

Full response: Reproduced verbatim in Section 3.2 above.

No other sections of this document used generative AI assistance.

7 Code Appendix

All code chunks used to produce this document are collected below. No code appears in the main body of the report.

```
library(ggplot2)
library(dplyr)
library(tidyr)
library(knitr)
library(kableExtra)
library(patchwork)
library(scales)

# ---- Busiest Airports Dataset (2020-2024) ----
# Six major US airports; passenger counts in millions
airports_wide <- data.frame(
  Airport = c("ATL", "DFW", "DEN", "ORD", "LAX", "JFK"),
  `2020` = c(42.9, 29.8, 33.1, 18.9, 26.9, 12.2),
  `2021` = c(75.7, 57.8, 58.8, 44.0, 48.0, 22.8),
  `2022` = c(93.6, 73.4, 69.3, 68.3, 65.9, 47.9),
  `2023` = c(104.7, 81.8, 77.8, 77.0, 75.0, 62.2),
  `2024` = c(108.0, 84.0, 82.0, 80.0, 79.0, 67.0),
  check.names = FALSE
)

airports_long <- airports_wide |>
  pivot_longer(cols = -Airport,
               names_to = "Year",
               values_to = "Passengers_Millions") |>
  mutate(Year = as.integer(Year))

# ---- Monte Carlo Integration Function ----
run_monte_carlo <- function(n, x_min, x_max, y_min, y_max, func) {
  x_coords <- runif(n, min = x_min, max = x_max)
  y_coords <- runif(n, min = y_min, max = y_max)
  flag     <- ifelse(y_coords <= func(x_coords), "on/below", "above")
  p        <- sum(flag == "on/below") / n
  rect_area <- (x_max - x_min) * (y_max - y_min)
  estimate  <- p * rect_area
  data.frame(x = x_coords, y = y_coords, flag = flag,
             n = n, estimate = estimate)
}

# Helper to build one subplot
make_mc_plot <- function(n) {
  sim <- run_monte_carlo(n = n, x_min = 0, x_max = 20,
                         y_min = 0, y_max = 0.16,
```

```r
                          func = function(x) dchisq(x, df = 5))
  est <- round(sim$estimate[1], 4)
  ggplot(sim, aes(x = x, y = y, color = flag)) +
    geom_point(alpha = 0.4, size = 0.7) +
    stat_function(fun = dchisq, args = list(df = 5),
                  xlim = c(0, 20), color = "darkblue", linewidth = 1.0) +
    scale_color_manual(values = c("above" = "#C0392B", "on/below" = "#27AE60")) +
    labs(title   = paste0("n = ", format(n, big.mark = ",")),
         subtitle = paste0("Estimate: ", est),
         x = "x", y = "y", color = NULL) +
    theme_minimal(base_size = 9) +
    theme(plot.title    = element_text(face = "bold", size = 9),
          plot.subtitle = element_text(color = "gray40", size = 7),
          legend.key.size = unit(0.35, "cm"),
          legend.text     = element_text(size = 7))
}

# ---- Calcium Dataset ----
raw <- read.csv("calcium.csv", check.names = FALSE)

# Columns 1-4 = Control group; columns 5-8 = Calcium supplement group
control   <- raw[, 1:4] |> setNames(c("Year 0","Year 1","Year 2","Year 3"))
treatment <- raw[, 5:8] |> setNames(c("Year 0","Year 1","Year 2","Year 3"))

control$id    <- seq_len(nrow(control))
treatment$id  <- seq_len(nrow(treatment))
control$Group <- "Control"
treatment$Group <- "Calcium"

calcium_long <- bind_rows(control, treatment) |>
  pivot_longer(cols = starts_with("Year"),
               names_to  = "Time",
               values_to = "Bone_Density") |>
  filter(!is.na(Bone_Density)) |>
  mutate(Time = factor(Time,
           levels = c("Year 0","Year 1","Year 2","Year 3")))
airports_wide |>
  arrange(desc(`2024`)) |>
  kbl(booktabs = TRUE, align = c("l", rep("r", 5)),
      col.names = c("Airport", "2020", "2021", "2022", "2023", "2024")) |>
  kable_styling(latex_options = c("striped", "hold_position"),
                full_width = FALSE) |>
  add_header_above(c(" " = 1, "Passengers (Millions)" = 5))
ggplot(airports_long, aes(x = Year, y = Passengers_Millions,
                          color = Airport, group = Airport)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
```

```r
  scale_x_continuous(breaks = 2020:2024) +
  scale_y_continuous(limits = c(0, 120),
                     labels = label_number(suffix = "M")) +
  scale_color_brewer(palette = "Set2") +
  labs(title    = "Passenger Traffic at Six Major Airports (2020-2024)",
       subtitle = "All six airports cratered in 2020 - recovery speed varied dramatically",
       x = "Year", y = "Total Passengers (Millions)", color = "Airport") +
  theme_minimal(base_size = 12) +
  theme(plot.title      = element_text(face = "bold"),
        plot.subtitle   = element_text(color = "gray40"),
        panel.grid.minor = element_blank(),
        legend.position = "right")
set.seed(184)
p10    <- make_mc_plot(10)
p100   <- make_mc_plot(100)
p1000  <- make_mc_plot(1000)
p10000 <- make_mc_plot(10000)

(p10 + p100) / (p1000 + p10000) +
  plot_annotation(
    title   = "Monte Carlo Integration: Chi-Square PDF (df = 5) over [0, 20]",
    theme   = theme(plot.title = element_text(face = "bold", size = 11))
  ) +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
raw2 <- read.csv("calcium.csv", check.names = FALSE)
colnames(raw2) <- c("Ctrl_Y0","Ctrl_Y1","Ctrl_Y2","Ctrl_Y3",
                    "Trt_Y0", "Trt_Y1", "Trt_Y2", "Trt_Y3")
raw2_long <- raw2 |>
  pivot_longer(everything(), names_to = "Variable", values_to = "Value") |>
  filter(!is.na(Value)) |>
  mutate(Variable = factor(Variable,
           levels = c("Ctrl_Y0","Ctrl_Y1","Ctrl_Y2","Ctrl_Y3",
                      "Trt_Y0","Trt_Y1","Trt_Y2","Trt_Y3")))

ggplot(raw2_long, aes(x = Variable, y = Value, fill = Variable)) +
  geom_boxplot(show.legend = FALSE) +
  labs(title = "Distribution of Calcium Dataset Variables",
       x = NULL, y = "Value") +
  scale_fill_brewer(palette = "Paired") +
  theme_minimal(base_size = 11) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
# Individual trajectories
means <- calcium_long |>
  group_by(Group, Time) |>
  summarise(mean_density = mean(Bone_Density), .groups = "drop")
```

```r
ggplot(calcium_long,
       aes(x = Time, y = Bone_Density, group = interaction(id, Group),
           color = Group)) +
  geom_line(alpha = 0.25, linewidth = 0.5) +
  geom_point(alpha = 0.3, size = 1.2) +
  geom_line(data = means,
            aes(x = Time, y = mean_density, group = Group, color = Group),
            linewidth = 2, inherit.aes = FALSE) +
  geom_point(data = means,
             aes(x = Time, y = mean_density, color = Group),
             size = 3.5, inherit.aes = FALSE) +
  scale_color_manual(values = c("Control" = "#2166AC", "Calcium" = "#D6604D")) +
  labs(title    = "Bone Density Trajectories: Control vs. Calcium Supplement",
       subtitle = "Thin lines = individual subjects; thick lines = group means",
       x = "Time Point", y = "Bone Density", color = "Group") +
  theme_minimal(base_size = 12) +
  theme(plot.title    = element_text(face = "bold"),
        plot.subtitle = element_text(color = "gray40"),
        panel.grid.minor = element_blank())
```